

PROJECT REPORT

VIT University

Department of VITyarthi(Fundamentals of AI and ML)

March 2026

Project:

Weather Predictor using python code

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Objective:

The objective of this project is to develop a predictive model that estimates **Ambient Temperature** based on three key environmental factors:

Humidity, Atmospheric Pressure, and Wind Speed. The project utilizes a Multiple Linear Regression approach to understand the linear relationship between these independent variables and the target output.

Results & Evaluation

Performance Metrics

- **The model was evaluated using standard regression metrics:**
- **Mean Absolute Error (MAE):** Measures the average magnitude of errors in the predictions.
- **Mean Squared Error (MSE):** Penalizes larger errors, providing a sense of outliers.

- **Predicted temperature:** $\text{Temperature} = 20 + 0.5 * \text{Humidity} - 0.02 * \text{Pressure} + 0.1 * \text{Wind Speed}$

Visualization of Results

A final scatter plot of **Actual vs. Predicted Temperature** was created. A strong diagonal trend in this plot indicates that the model's predictions closely align with the true values, confirming the effectiveness of the linear regression approach for this dataset.

Conclusion

The model successfully identifies the underlying patterns in the weather data. While this implementation uses synthetic data, the pipeline is fully compatible with real-world meteorological datasets. Future iterations could involve **Polynomial Regression** to capture non-linear weather patterns or **Feature Scaling** to handle the different units of pressure and humidity.